Chapter 14:
Polymer Structures

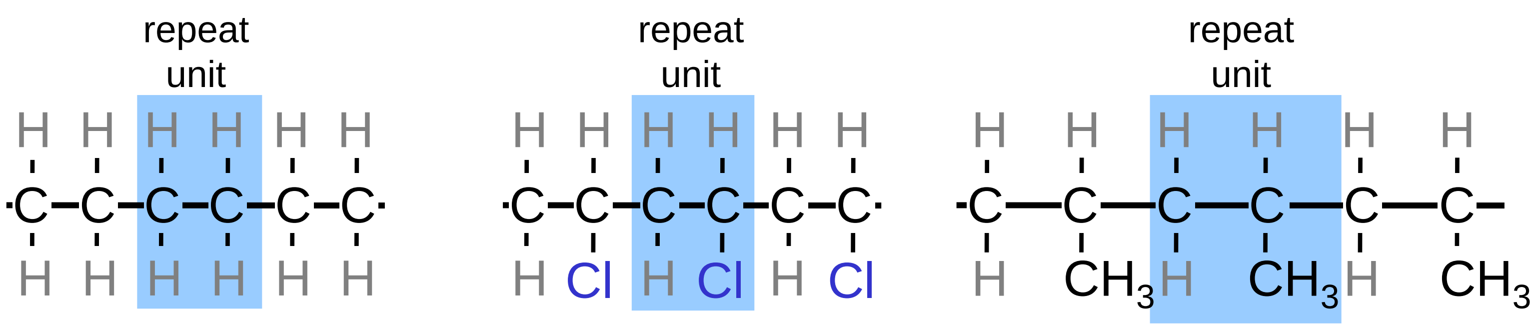
ISSUES TO ADDRESS...

- What are the general structural and chemical characteristics of polymer molecules?
- What are some of the common polymeric materials, and how do they differ chemically?
- How is the crystalline state in polymers different from that in metals and ceramics ?

What is a Polymer?

Poly mer

many repeat unit



Polyethylene (PE) Poly(vinyl chloride) (PVC) Polypropylene (PP)

Adapted from Fig. 14.2, *Callister & Rethwisch 8e.*

Ancient Polymers

- Originally natural polymers were used
  – Wood
  – Cotton
  – Leather
  – Rubber
  – Wool
  – Silk

- Oldest known uses
  – Rubber balls used by Incas
  – Noah used pitch (a natural polymer) for the ark

Polymer Composition

Most polymers are hydrocarbons
– i.e., made up of H and C

- Saturated hydrocarbons
  - Each carbon singly bonded to four other atoms
  - Example:
    - Ethane, C_2H_6

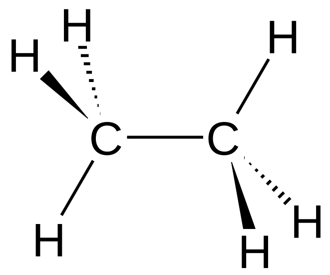
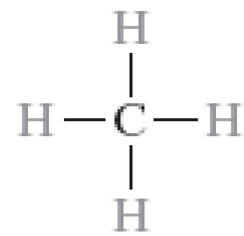
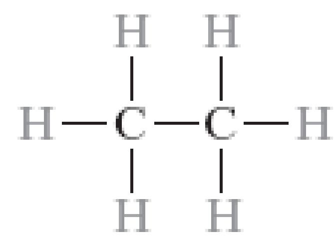
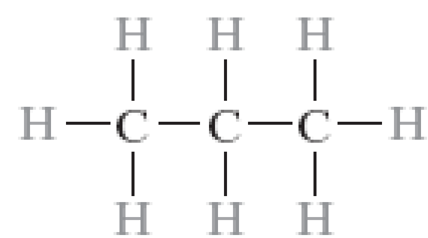
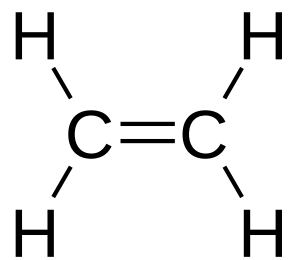


Table 14.1 **Compositions and Molecular Structures for Some of the Paraffin Compounds: C_nH_{2n+2}**

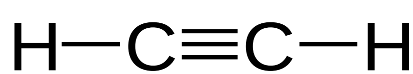
Name	*Composition*	*Structure*	*Boiling Point (°C)*
Methane	CH_4		−164
Ethane	C_2H_6		−88.6
Propane	C_3H_8		−42.1
Butane	C_4H_{10}		−0.5
Pentane	C_5H_{12}		36.1
Hexane	C_6H_{14}		69.0

Unsaturated Hydrocarbons

- Double & triple bonds somewhat unstable – can form new bonds
  - Double bond found in ethylene or ethene - C_2H_4



  - Triple bond found in acetylene or ethyne - C_2H_2

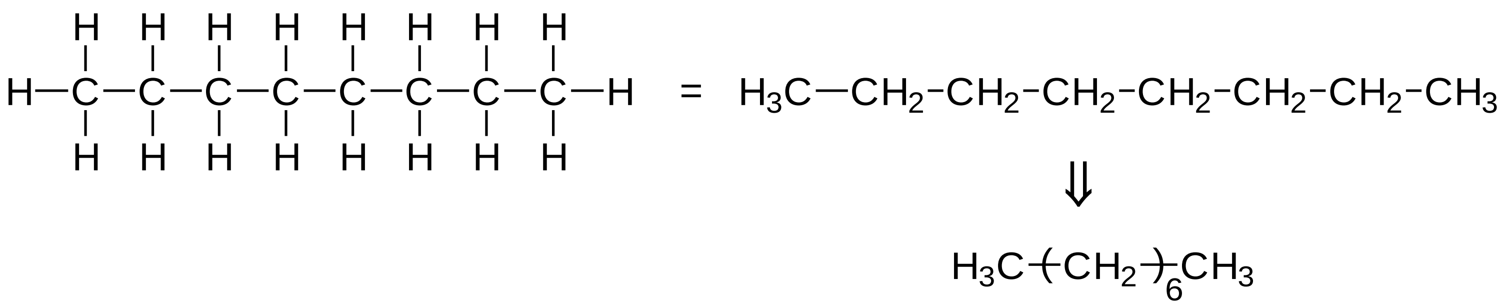


Isomerism

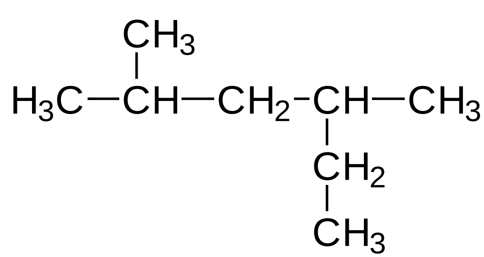
- Isomerism
  - two compounds with same chemical formula can have quite different structures

    for example: C_8H_{18}

    - normal-octane

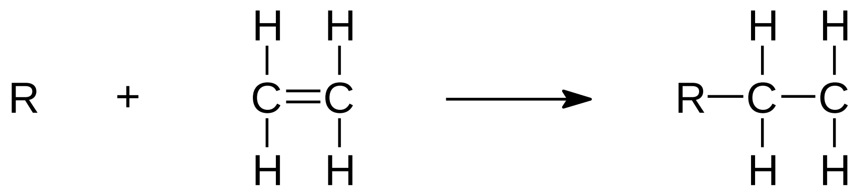


    - 2,4-dimethylhexane



Polymerization and Polymer Chemistry

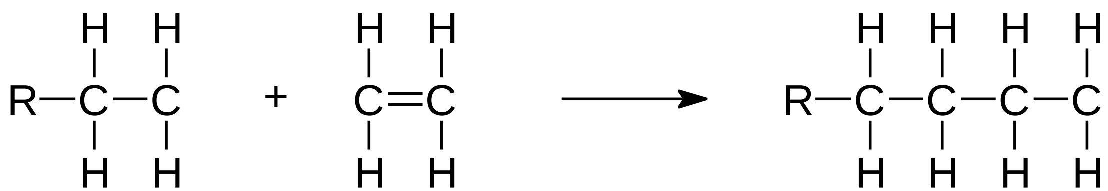
- Free radical polymerization



initiation

free radical

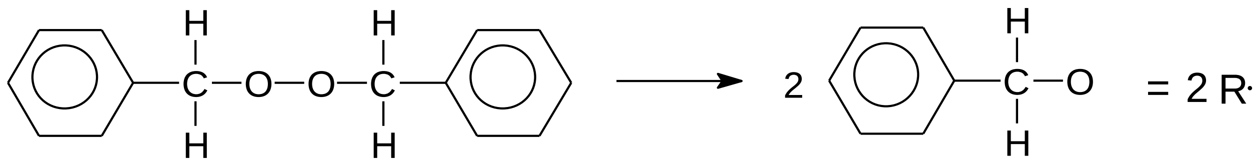
monomer (ethylene)



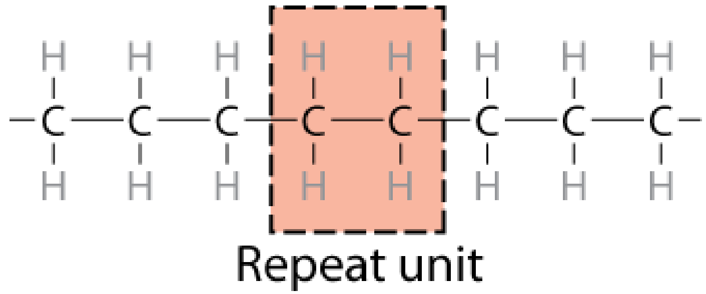
propagation

dimer

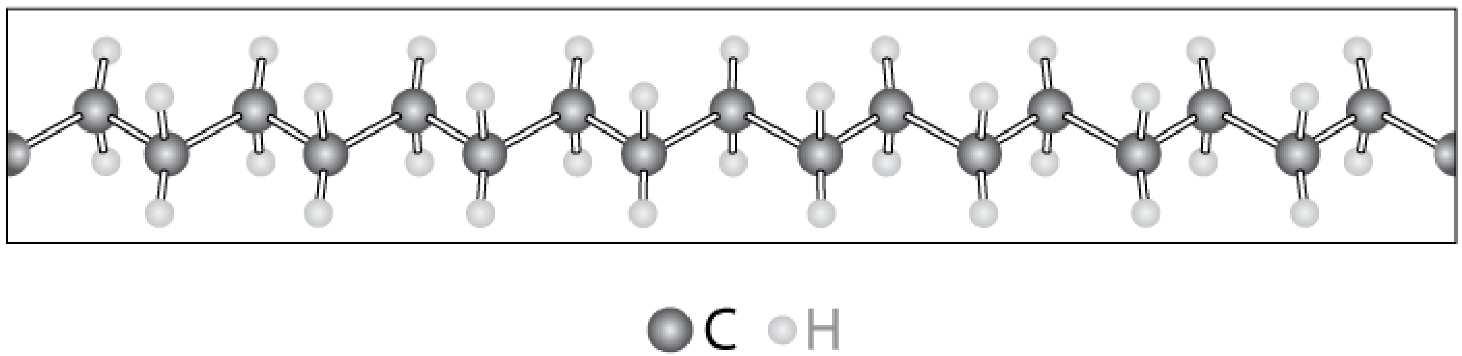
- Initiator: example - benzoyl peroxide



Chemistry and Structure of Polyethylene



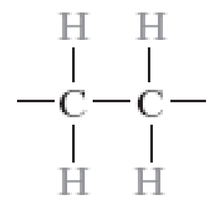
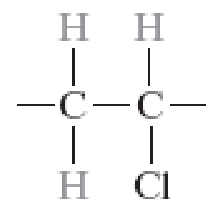
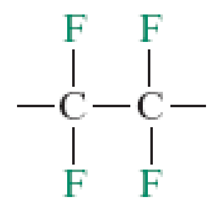
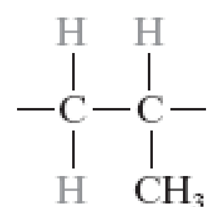
Adapted from Fig. 14.1, *Callister & Rethwisch 8e.*



Note: polyethylene is a long-chain hydrocarbon
- paraffin wax for candles is short polyethylene

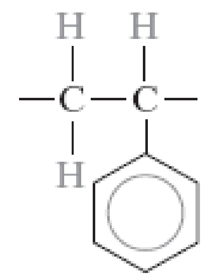
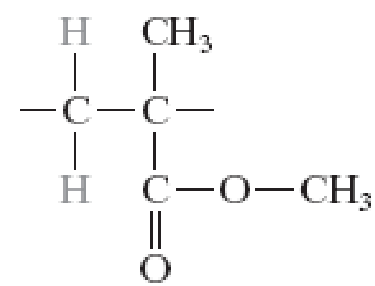
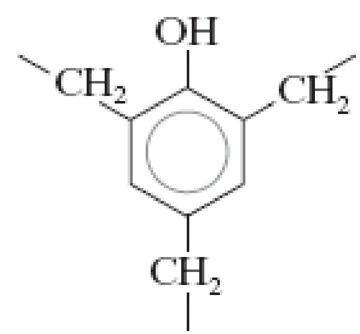
Bulk or Commodity Polymers

Table 14.3 **A Listing of Repeat Units for 10 of the More Common Polymeric Materials**

Polymer	*Repeat Unit*
Polyethylene (PE)	
Poly(vinyl chloride) (PVC)	
Polytetrafluoroethylene (PTFE)	
Polypropylene (PP)	

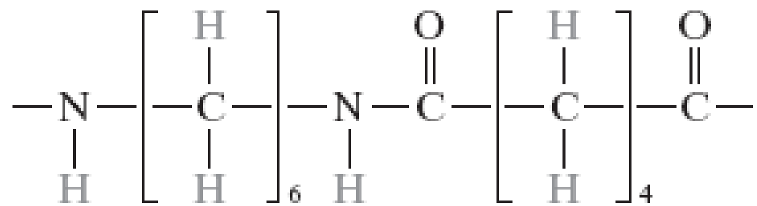
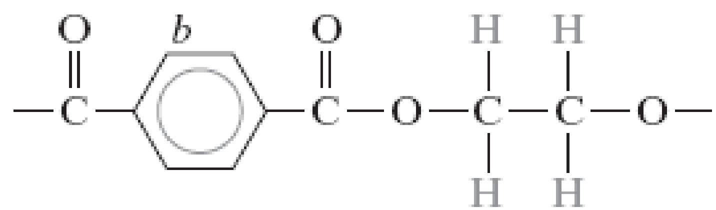
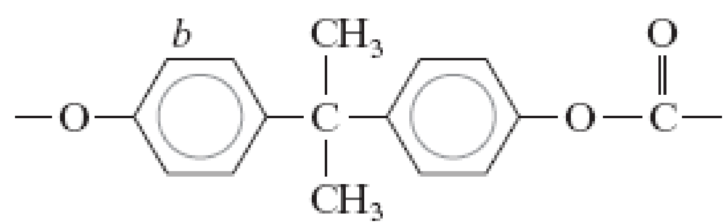
Bulk or Commodity Polymers (cont)

Table 14.3 **A Listing of Repeat Units for 10 of the More Common Polymeric Materials**

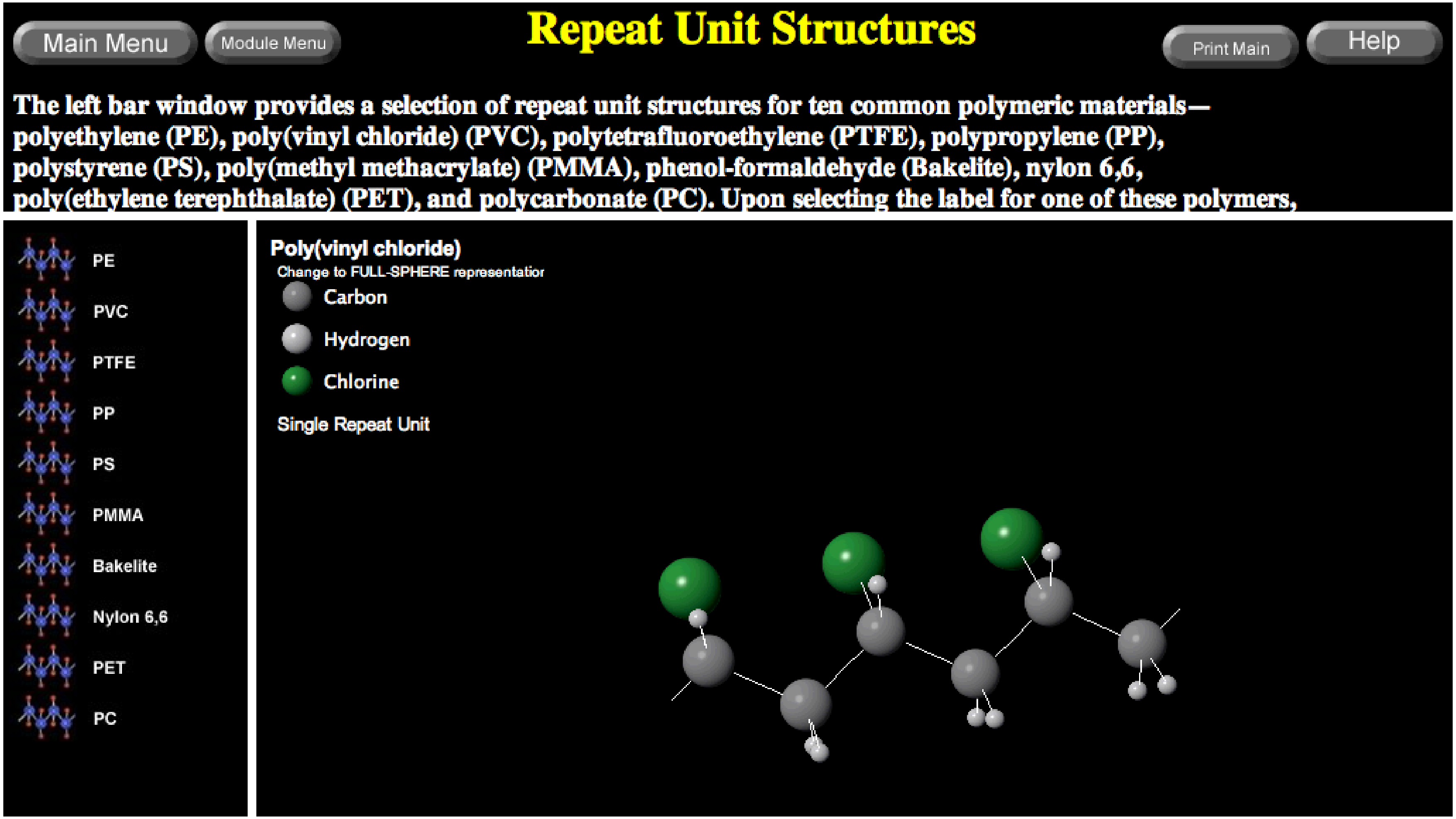
Polymer	*Repeat Unit*
Polystyrene (PS)	
Poly(methyl methacrylate) (PMMA)	
Phenol-formaldehyde (Bakelite)	

Bulk or Commodity Polymers (cont)

Table 14.3 **A Listing of Repeat Units for 10 of the More Common Polymeric Materials**

Polymer	*Repeat Unit*
Poly(hexamethylene adipamide) (nylon 6,6)	
Poly(ethylene terephthalate) (PET, a polyester)	
Polycarbonate (PC)	

VMSE: Polymer Repeat Unit Structures

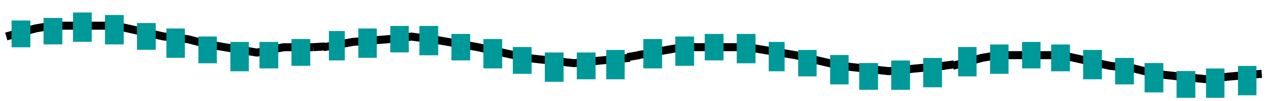


Manipulate and rotate polymer structures in 3-dimensions

MOLECULAR WEIGHT

- Molecular weight, M: Mass of a mole of chains.

Low M



high M

Not all chains in a polymer are of the same length

— i.e., there is a distribution of molecular weights

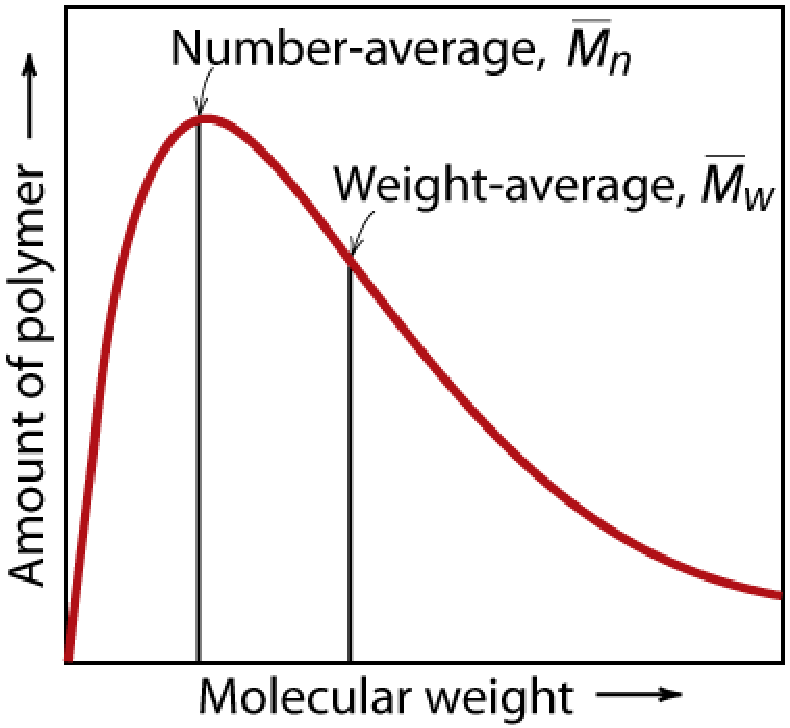
MOLECULAR WEIGHT DISTRIBUTION

Adapted from Fig. 14.4, *Callister & Rethwisch 8e.*

$$\overline{M}_n = \frac{\text{total wt of polymer}}{\text{total \# of molecules}}$$

$$\overline{M}_n = \Sigma x_i M_i$$

$$\overline{M}_w = \Sigma w_i M_i$$



M_i = mean (middle) molecular weight of size range *i*

x_i = number fraction of chains in size range *i*

w_i = weight fraction of chains in size range *i*

Molecular Weight Calculation

Example: average mass of a class

Student	Weight mass (lb)
1	104
2	116
3	140
4	143
5	180
6	182
7	191
8	220
9	225
10	380

What is the average weight of the students in this class:

a) Based on the number fraction of students in each mass range?
b) Based on the weight fraction of students in each mass range?

Molecular Weight Calculation (cont.)

Solution: The first step is to sort the students into weight ranges. Using 40 lb ranges gives the following table:

weight range mass (lb)	number of students N_i	mean weight W_i mass (lb)
81-120	2	110
121-160	2	142
161-200	3	184
201-240	2	223
241-280	0	-
281-320	0	-
321-360	0	-
361-400	1	380

total number → ΣN_i 10 $\Sigma N_i W_i$ 1881 ← total weight

Calculate the number and weight fraction of students in each weight range as follows:

$$x_i = \frac{N_i}{\sum N_i} \qquad w_i = \frac{N_i W_i}{\sum N_i W_i}$$

For example: for the 81-120 lb range

$$x_{81-120} = \frac{2}{10} = 0.2$$

$$w_{81-120} = \frac{2 \times 110}{1881} = 0.117$$

Molecular Weight Calculation (cont.)

weight range mass (lb)	mean weight M_i mass (lb)	number fraction x_i	weight fraction w_i
81-120	110	0.2	0.117
121-160	142	0.2	0.150
161-200	184	0.3	0.294
201-240	223	0.2	0.237
241-280	-	0	0.000
281-320	-	0	0.000
321-360	-	0	0.000
361-400	380	0.1	0.202

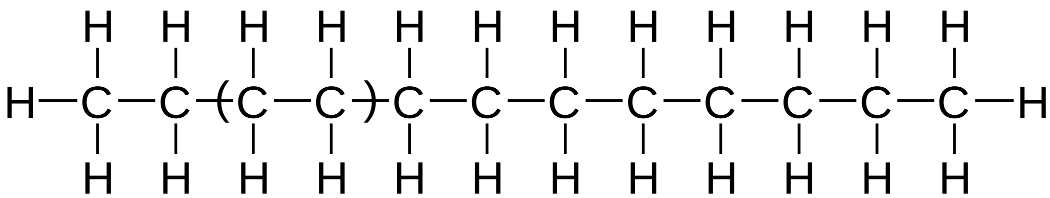
$$\boxed{\overline{M}_n = \sum x_i M_i} = (0.2 \text{ x } 110 + 0.2 \text{ x } 142 + 0.3 \text{ x } 184 + 0.2 \text{ x } 223 + 0.1 \text{ x } 380) = \boxed{188 \text{ lb}}$$

$$\boxed{\overline{M}_w = \sum w_i M_i} = (0.117 \text{ x } 110 + 0.150 \text{ x } 142 + 0.294 \text{ x } 184 + 0.237 \text{ x } 223 + 0.202 \text{ x } 380) = \boxed{218 \text{ lb}}$$

$$\overline{M}_w = \sum w_i M_i = 218 \text{ lb}$$

Degree of Polymerization, *DP*

DP = average number of repeat units per chain



$DP = 6$

$$DP = \frac{\overline{M}_n}{\overline{m}}$$

where $\overline{m}$ = average molecular weight of repeat unit

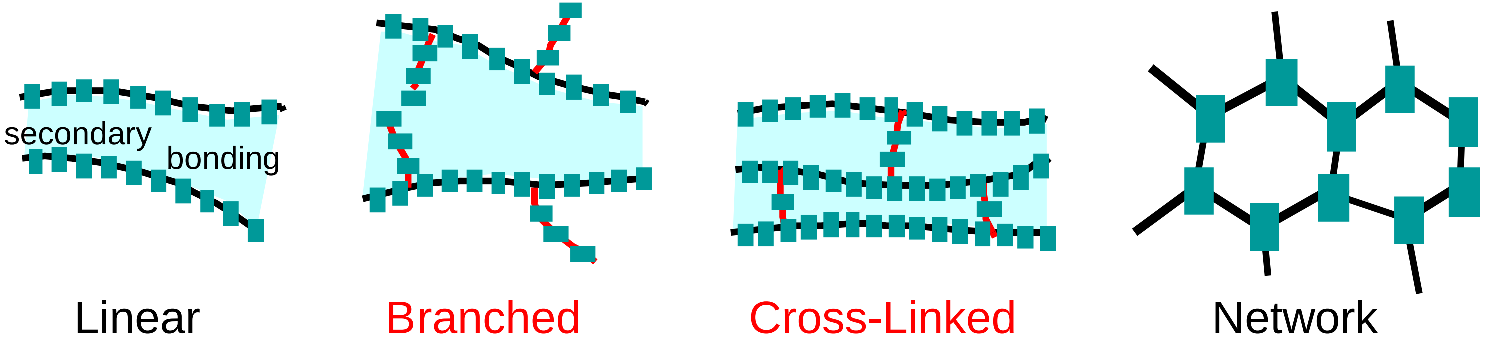
for copolymers this is calculated as follows:

$$\overline{m} = \Sigma f_i m_i$$

Chain fraction

mol. wt of repeat unit *i*

Molecular Structures for Polymers

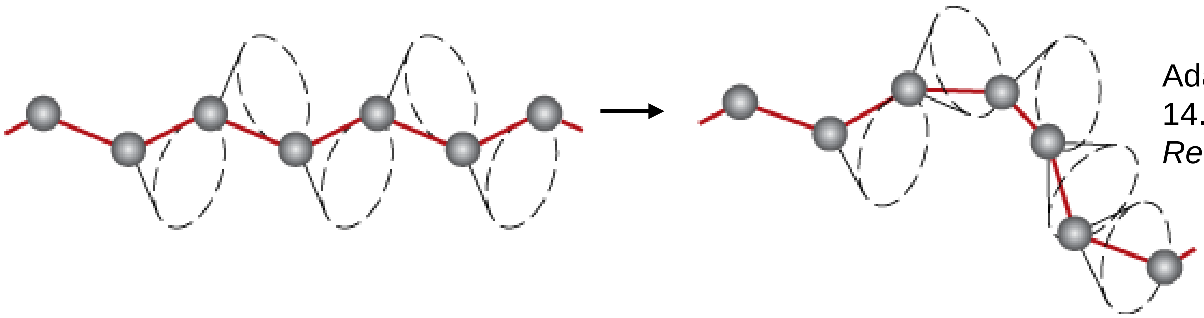


Adapted from Fig. 14.7, *Callister & Rethwisch 8e.*

Polymers – Molecular Shape

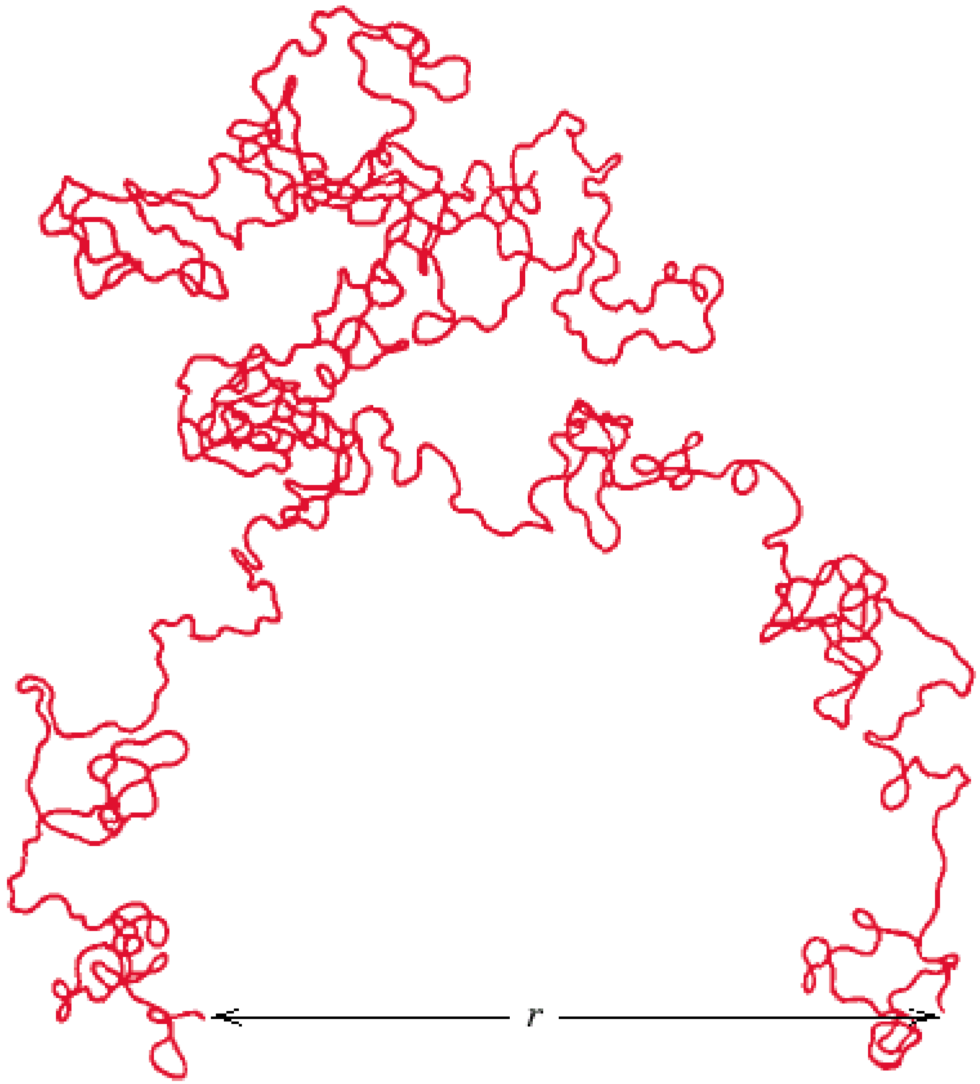
Molecular Shape (or Conformation) – chain bending and twisting are possible by rotation of carbon atoms around their chain bonds

- note: not necessary to break chain bonds to alter molecular shape



Adapted from Fig. 14.5, *Callister & Rethwisch 8e.*

Chain End-to-End Distance, *r*

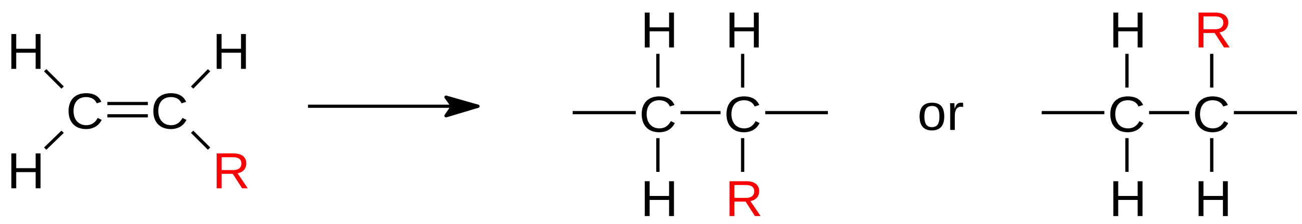


Adapted from Fig. 14.6, *Callister & Rethwisch 8e.*

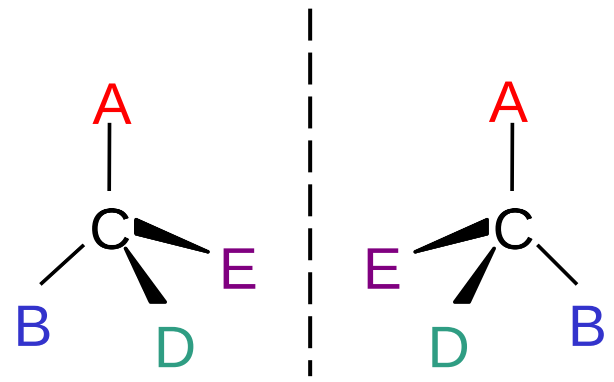
Molecular Configurations for Polymers

Configurations – to change must break bonds

- Stereoisomerism



Stereoisomers are mirror images – can't superimpose without breaking a bond



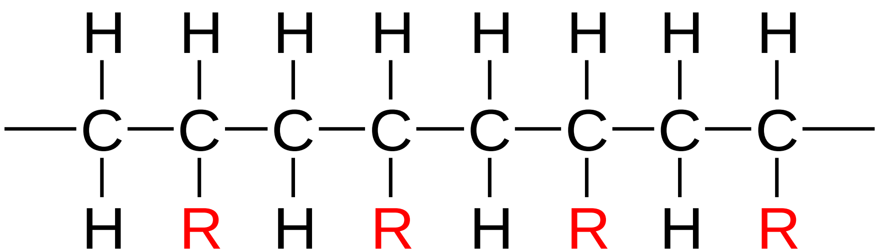
mirror plane

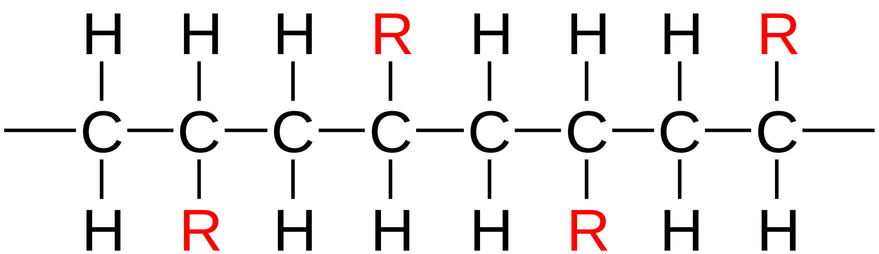
Tacticity

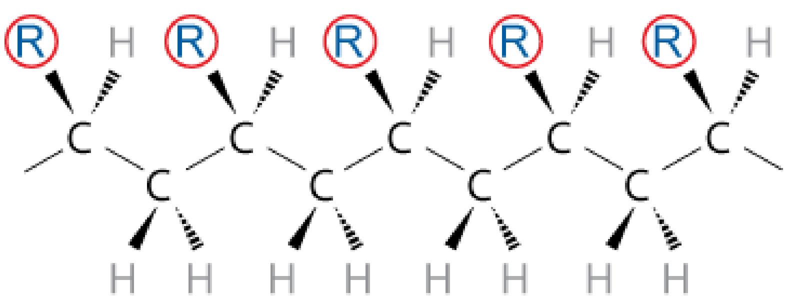
Tacticity – stereoregularity or spatial arrangement of R units along chain

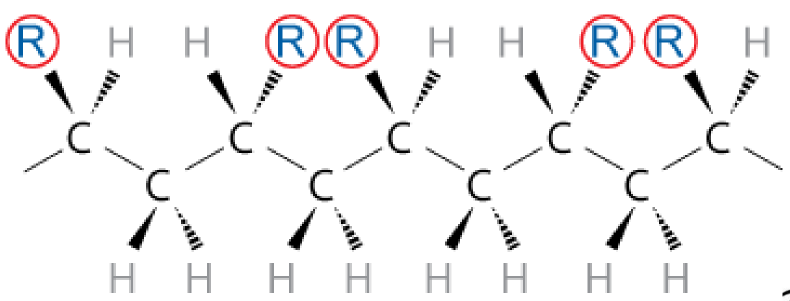
isotactic – all R groups on same side of chain

syndiotactic – R groups alternate sides



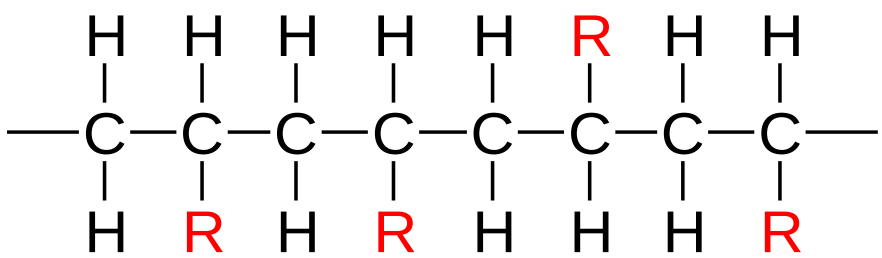


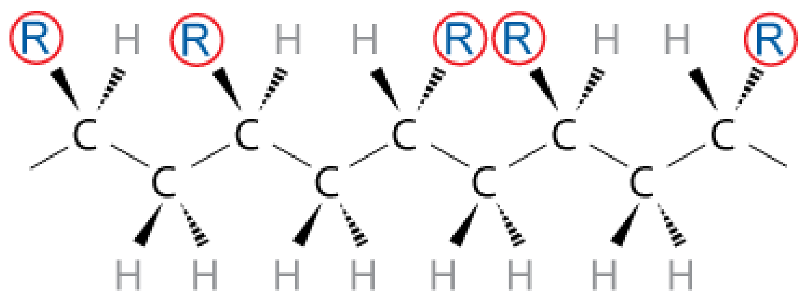




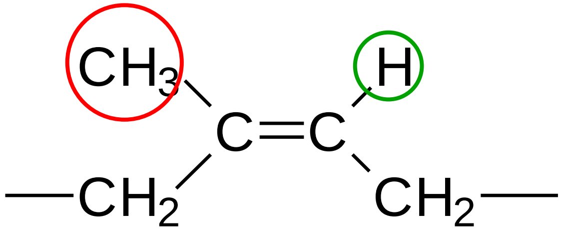
Tacticity (cont.)

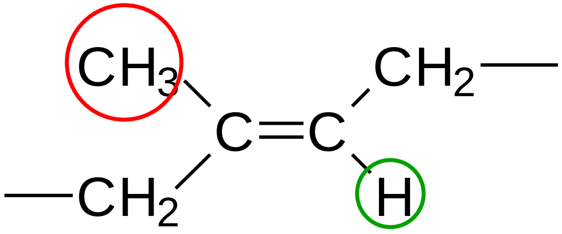
atactic – R groups randomly positioned





cis/trans Isomerism





cis

cis-isoprene
(natural rubber)

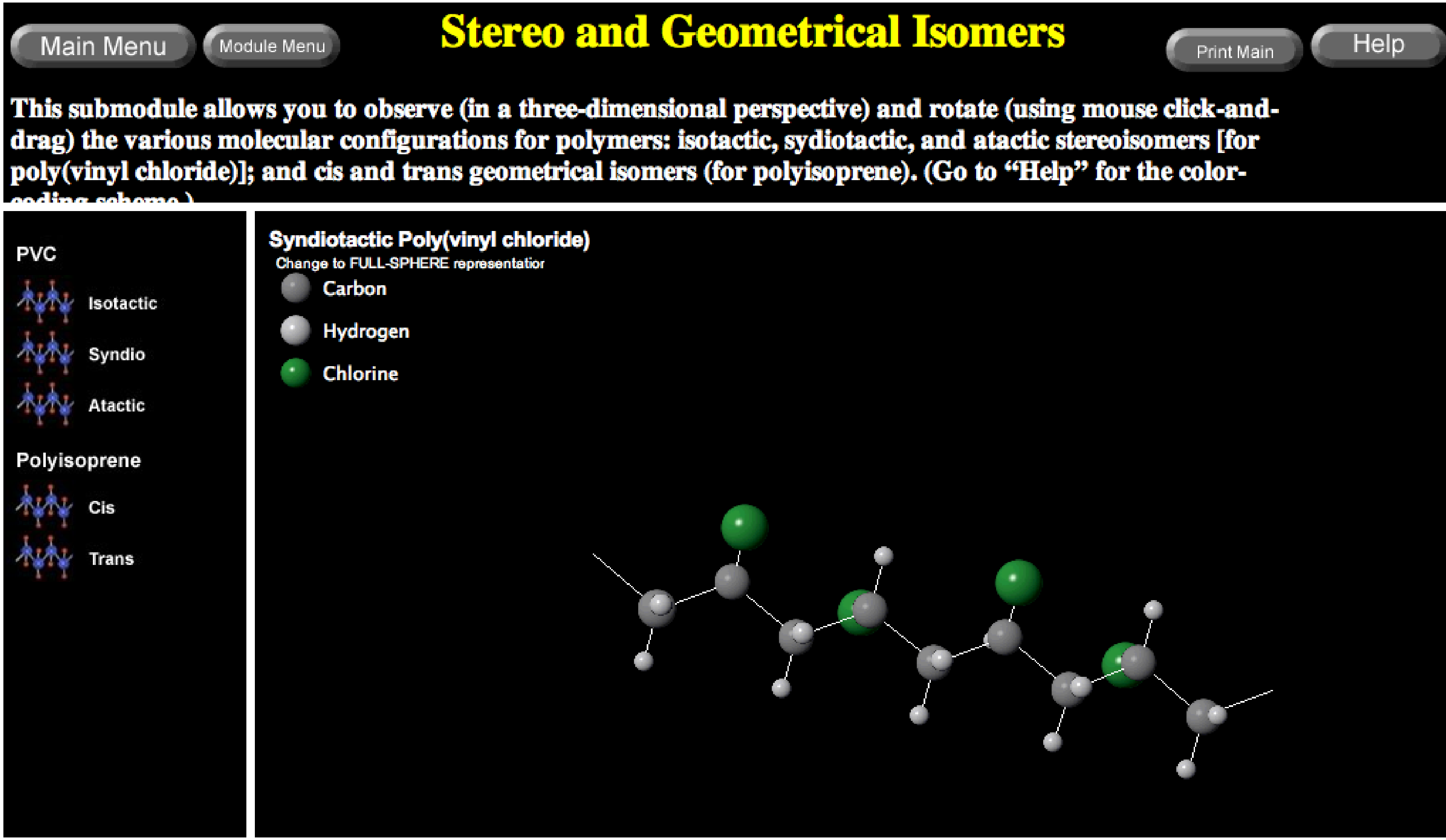
H atom and CH_3 group on same side of chain

trans

trans-isoprene
(gutta percha)

H atom and CH_3 group on opposite sides of chain

VMSE: Stereo and Geometrical Isomers



Manipulate and rotate polymer structures in 3-dimensions

Copolymers

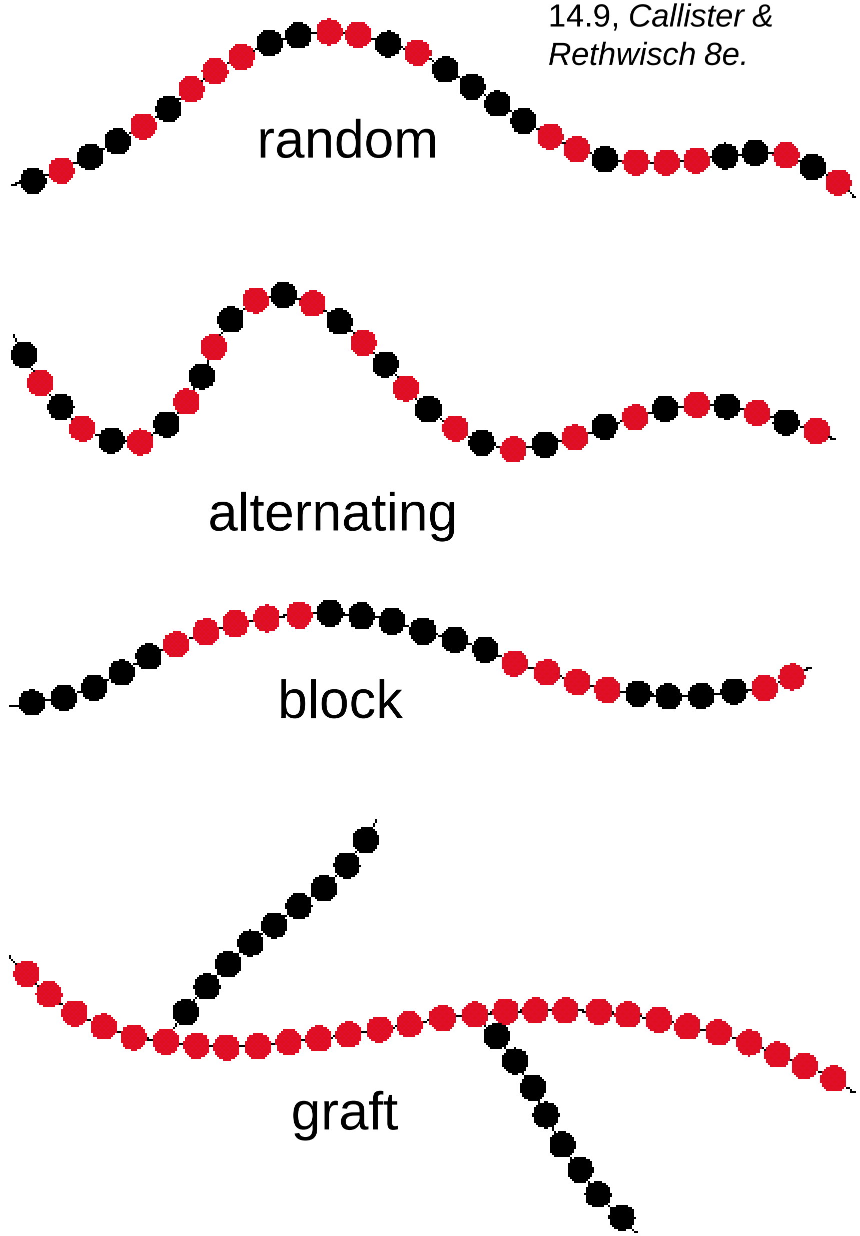
Adapted from Fig. 14.9, *Callister & Rethwisch 8e.*

two or more monomers polymerized together

- random – A and B randomly positioned along chain
- alternating – A and B alternate in polymer chain
- block – large blocks of A units alternate with large blocks of B units
- graft – chains of B units grafted onto A backbone

A –

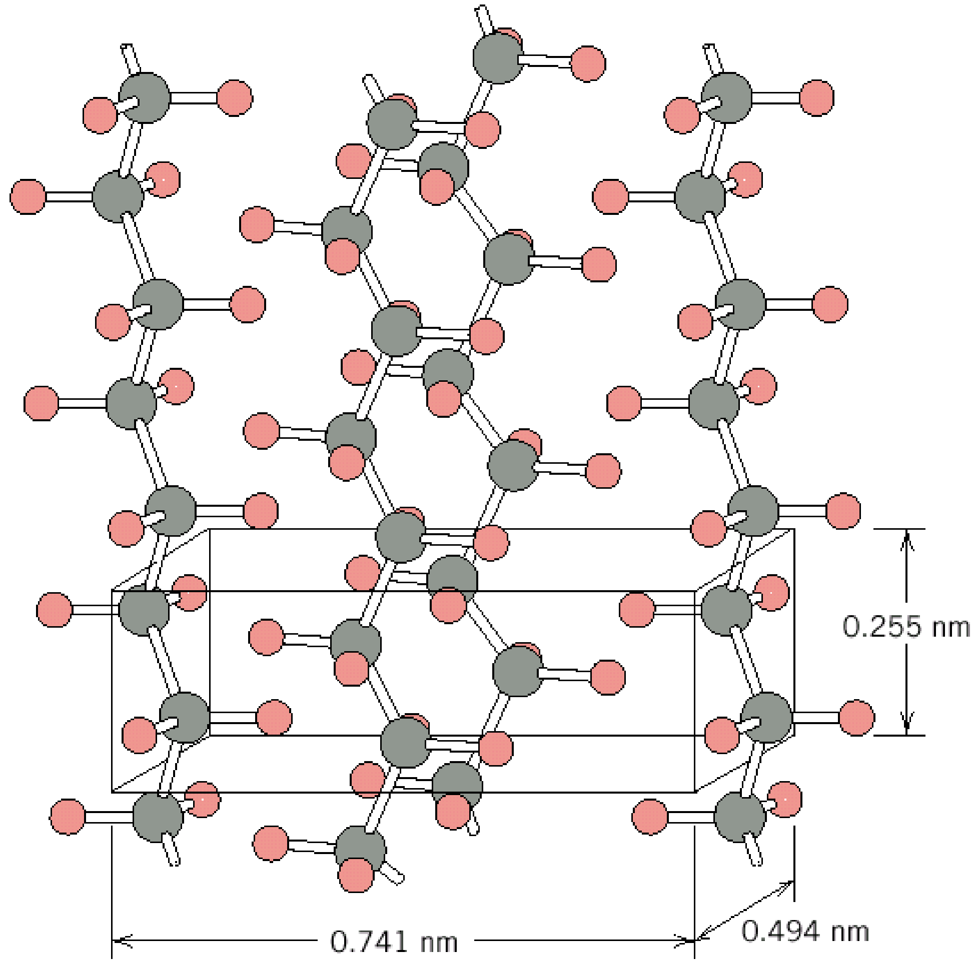
B –



Crystallinity in Polymers

Adapted from Fig. 14.10, *Callister & Rethwisch 8e.*

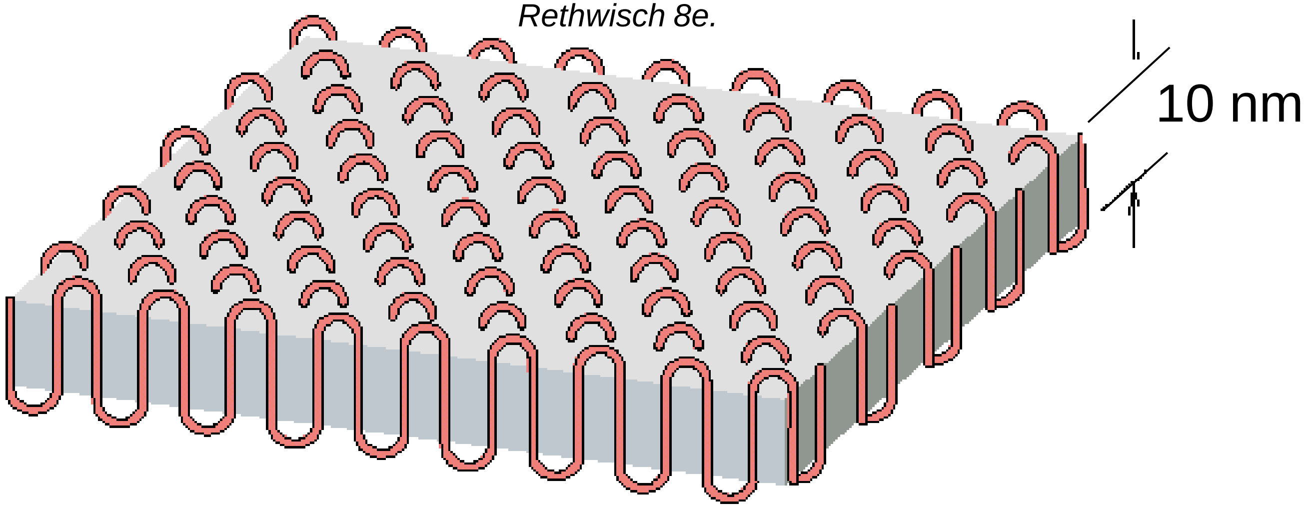
- Ordered atomic arrangements involving molecular chains
- Crystal structures in terms of unit cells
- Example shown
  - – polyethylene unit cell



Polymer Crystallinity

- Crystalline regions
  - thin platelets with chain folds at faces
  - Chain folded structure

Adapted from Fig. 14.12, *Callister & Rethwisch 8e.*

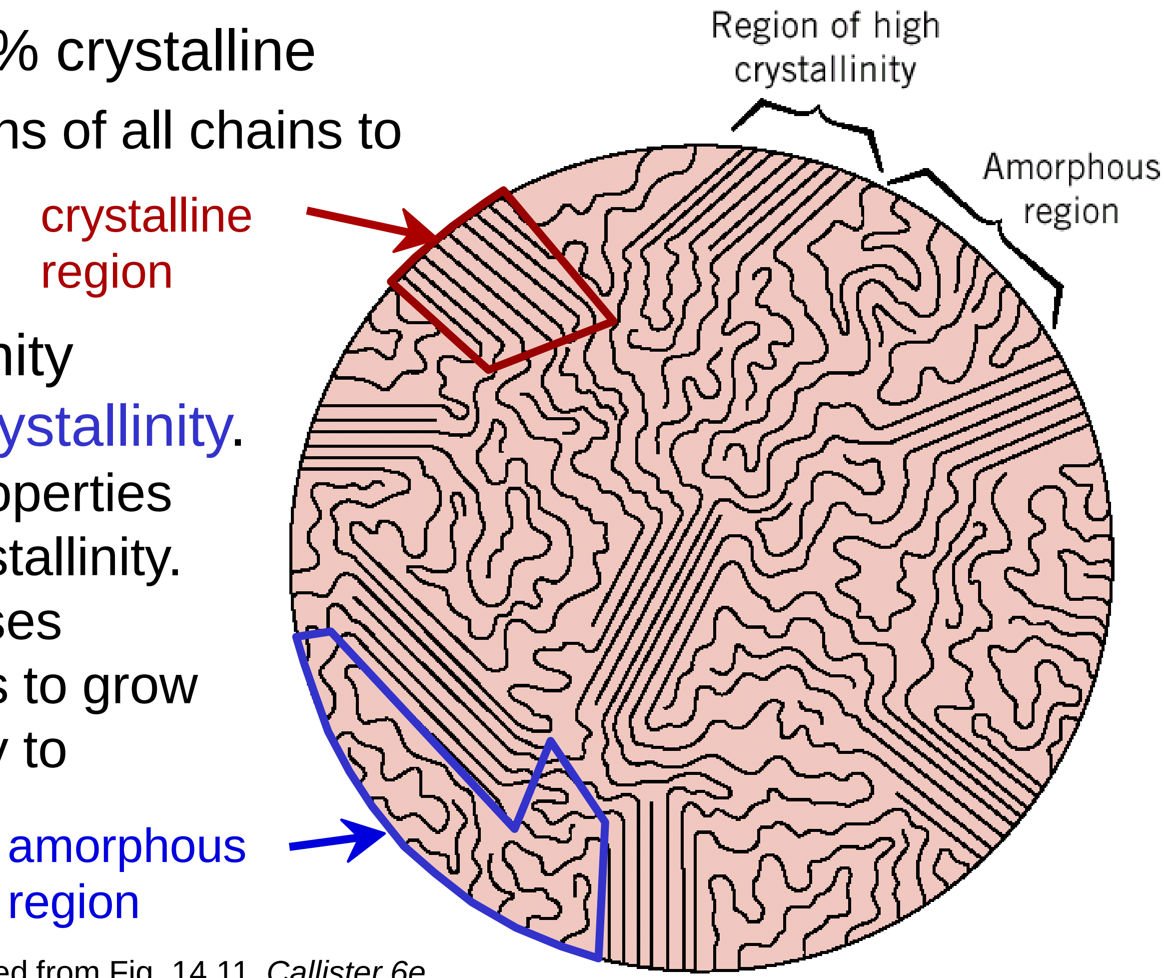


Polymer Crystallinity (cont.)

Polymers rarely 100% crystalline

- Difficult for all regions of all chains to become aligned

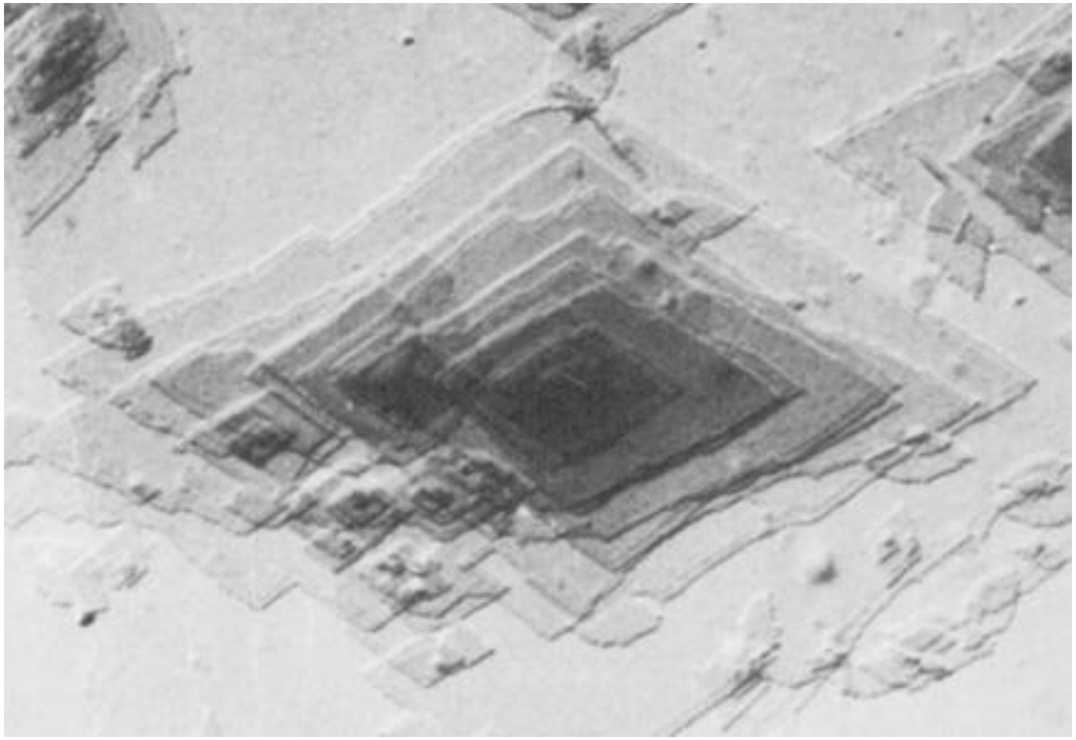
- Degree of crystallinity expressed as % crystallinity.
  -- Some physical properties depend on % crystallinity.
  -- Heat treating causes crystalline regions to grow and % crystallinity to increase.



Adapted from Fig. 14.11, *Callister 6e.*
(Fig. 14.11 is from H.W. Hayden, W.G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*, John Wiley and Sons, Inc., 1965.)

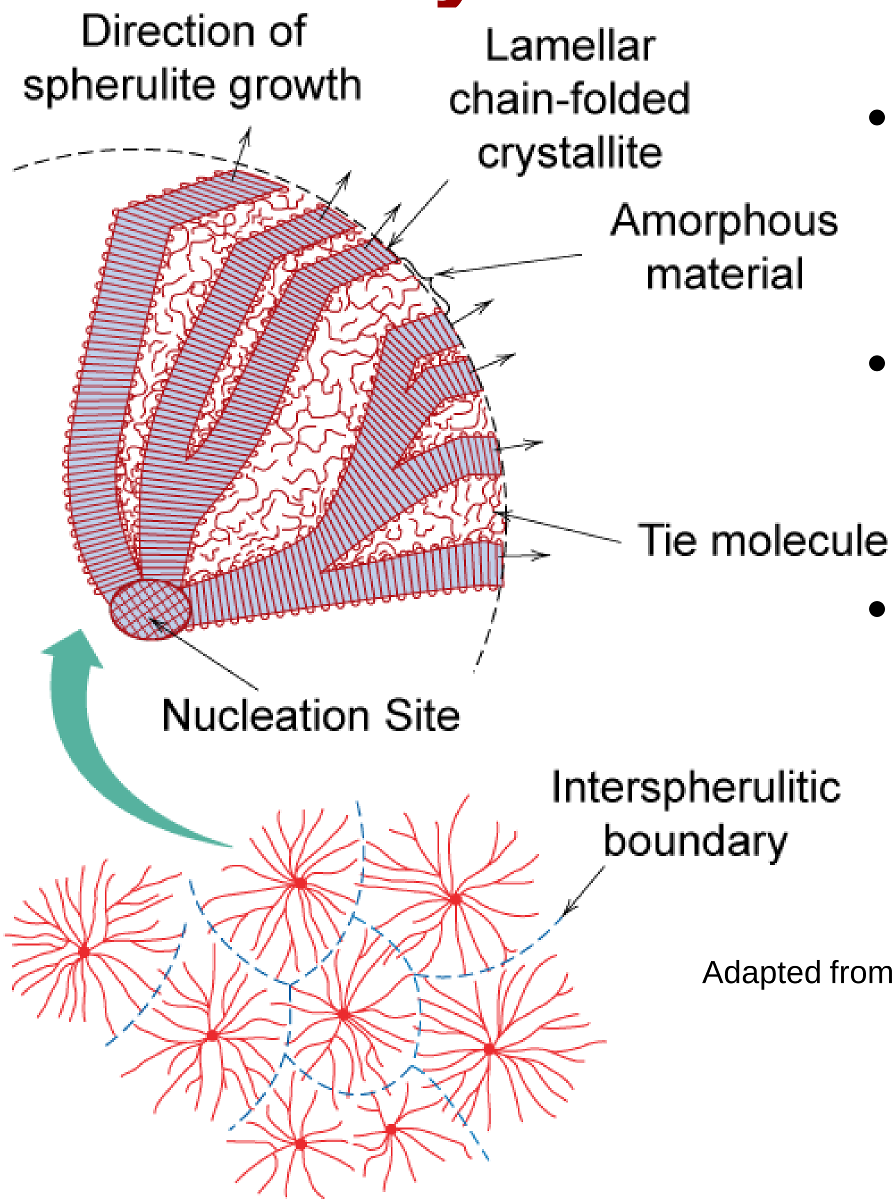
Polymer Single Crystals

- Electron micrograph – multilayered single crystals (chain-folded layers) of polyethylene
- Single crystals – only for slow and carefully controlled growth rates



Adapted from Fig. 14.11, *Callister & Rethwisch 8e.*

Semicrystalline Polymers



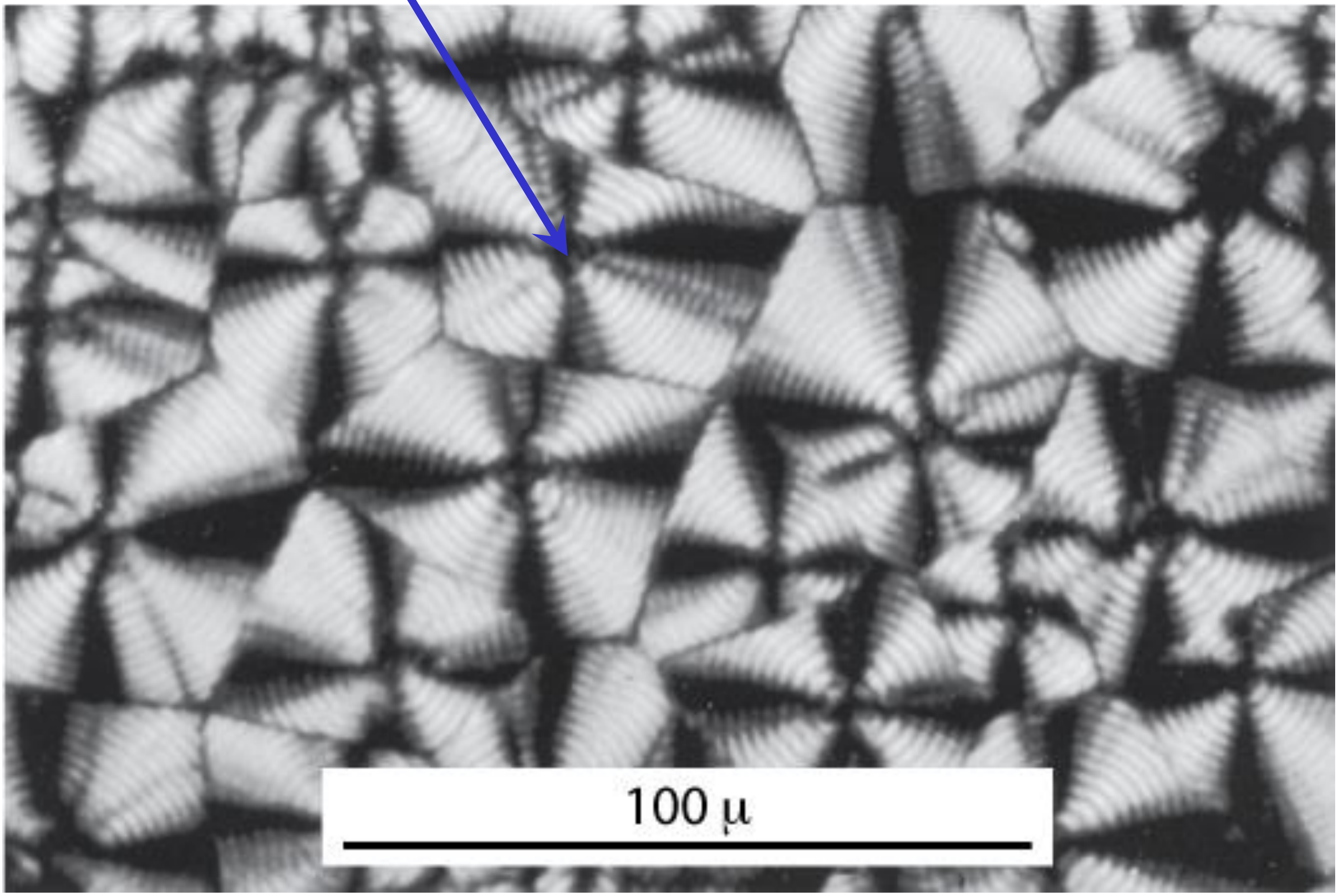
- Some semicrystalline polymers form spherulite structures
- Alternating chain-folded crystallites and amorphous regions
- Spherulite structure for relatively rapid growth rates

Adapted from Fig. 14.13, *Callister & Rethwisch 8e.*

Photomicrograph – Spherulites in Polyethylene

Cross-polarized light used

-- a maltese cross appears in each spherulite



Adapted from Fig. 14.14, *Callister & Rethwisch 8e.*

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: